Entropy and energetics

Why does sodium chloride exist, why is it $CaCl_2$ and not CaCl, and why do some reactions that take in heat still happen by themselves? This topic answers with two tools: Born–Haber and solution cycles, which measure the energy holding ionic solids together, and entropy, which counts the ways energy can be shared out and decides whether a change is feasible at all.

The enthalpy changes behind an ionic lattice

The **lattice energy** of an ionic compound measures how strongly its ions hold each other. It cannot be measured directly, because you cannot collect a mole of free gaseous ions and let them crash together. Instead it is calculated with Hess's law from enthalpy changes that *can* be measured. You need to know each of them precisely. All are standard values: measured at 100 kPa and a stated temperature, usually 298 K, with substances in their standard states.

Enthalpy change	Definition	Example	Sign
Enthalpy change of formation, $\Delta_f H$	1 mol of a compound forms from its elements in their standard states	$Na(s) + \frac{1}{2}Cl_2(g) \rightarrow NaCl(s)$	usually −
Enthalpy change of atomisation, $\Delta_{at} H$	1 mol of **gaseous atoms** forms from the element in its standard state	$Na(s) \rightarrow Na(g)$ $\frac{1}{2}Cl_2(g) \rightarrow Cl(g)$	always +
First ionisation energy	1 mol of electrons is removed from 1 mol of gaseous atoms, giving 1 mol of gaseous 1+ ions	$Mg(g) \rightarrow Mg^+(g) + e^-$	always +
Second ionisation energy	1 mol of electrons is removed from 1 mol of gaseous 1+ ions, giving 1 mol of gaseous 2+ ions	$Mg^+(g) \rightarrow Mg^{2+}(g) + e^-$	always +
First electron affinity	1 mol of gaseous atoms each gain one electron, giving 1 mol of gaseous 1− ions	$Cl(g) + e^- \rightarrow Cl^-(g)$	− for these elements
Second electron affinity	1 mol of gaseous 1− ions each gain one electron, giving 1 mol of gaseous 2− ions	$O^-(g) + e^- \rightarrow O^{2-}(g)$	+
Lattice energy, LE	1 mol of an ionic solid forms from its **gaseous ions**	$Na^+(g) + Cl^-(g) \rightarrow NaCl(s)$	always −

- **Why is the first electron affinity negative?** The incoming electron is attracted by the nucleus of the neutral atom, so energy is released. Going down group 7 from chlorine to iodine it becomes less exothermic: the atoms are bigger, with more shielding, so the added electron is held less strongly.
- **Why is the second electron affinity positive?** The electron is being added to an ion that is already negative. Energy has to be put in to overcome the repulsion between them. For oxygen it is +798 kJ mol^{-1}, much larger than the −141 kJ mol^{-1} released by the first.
- **Atomisation of a diatomic element** is per mole of atoms formed, so $\Delta_{at}H(Cl)$ is half the Cl–Cl bond enthalpy: about +122 kJ mol^{-1} from a bond enthalpy of about +243 kJ mol^{-1}.

COMMON MISTAKE

Lattice energy is defined here as the enthalpy change for **forming** the solid from gaseous ions, so it is negative. The reverse process, breaking the lattice apart, has the same size but a positive sign. Read every question carefully to see which direction it means, and always write state symbols: the ions must be gaseous.

Born–Haber cycles

A Born–Haber cycle is a Hess's law cycle that links the formation of an ionic solid directly from its elements with an indirect route through gaseous atoms and gaseous ions. Both routes start and finish in the same place, so their total enthalpy changes are equal:

Born–Haber cycle for a compound MX

$$\Delta_f H = \Delta_{at} H(\mathrm{M}) + \mathrm{IE}(\mathrm{M}) + \Delta_{at} H(\mathrm{X}) + \mathrm{EA}(\mathrm{X}) + \mathrm{LE}(\mathrm{MX})$$

IE: ionisation energy (or energies) EA: electron affinity (or affinities) LE: lattice energy

Lattice energy of sodium chloride

1. Data / kJ mol^{-1}: $\Delta_f H(NaCl) = -411$; $\Delta_{at}H(Na) = +107$; first IE(Na) = +496; $\Delta_{at}H(Cl) = +122$; first EA(Cl) = −349.

2. Go round the indirect route from the elements to the gaseous ions: atomise sodium, ionise it, atomise chlorine, add the electron to chlorine.

$$107 + 496 + 122 + (-349) = +376 \ \mathrm{kJ\,mol^{-1}}$$

So $Na^+(g)$ + $Cl^-(g)$ sits 376 kJ mol^{-1} above the elements.

3 Hess's law: $\Delta_f H = 376 + \text{LE}$, so

$$\text{LE} = \Delta_f H - 376 = -411 - 376 = -787 \text{ kJ mol}^{-1}$$

- **Group 2 halides such as $MgCl_2$:** include both the first and second ionisation energies of the metal, and **twice** the atomisation enthalpy and electron affinity of chlorine, because there are two chloride ions. ($2 \times \Delta_{at}H(Cl)$ equals the Cl–Cl bond enthalpy.)
- **Oxides such as MgO:** include the first *and second* electron affinities of oxygen. The second is endothermic, so its arrow points **up** in the cycle.
- Any one value in the cycle can be found if all the others are known: exam questions often ask for an electron affinity or an atomisation enthalpy instead of the lattice energy.

EXAM TIP

When drawing a cycle, put the gaseous ions at the top, with every species and state symbol on each level, and include the electrons (e^-) on the levels where they are free. Up arrows are endothermic and down arrows exothermic.

Why calcium chloride is $CaCl_2$

Born–Haber cycles can test compounds that don't exist. A theoretical lattice energy can be estimated for CaCl and $CaCl_3$ from the sizes and charges of the ions, and the cycle then predicts their enthalpies of formation.

Compound	Ionisation energies needed / kJ mol^{-1}	Lattice energy / kJ mol^{-1}	$\Delta_f H$ / kJ mol^{-1}
CaCl	590	about −720 (estimated)	about −180
$CaCl_2$	590 + 1145 = 1735	−2255 (from the Born–Haber cycle)	−796 (measured)
$CaCl_3$	590 + 1145 + 4912 = 6647	about −4600 (estimated)	about +1540

The enthalpy of formation is largely a contest between the energy needed to make the cation and the energy released when the lattice forms. Going from Ca^+ to Ca^{2+} costs an extra 1145 kJ mol^{-1}, but the doubly charged ion attracts two chloride ions far more strongly, and the lattice energy grows by over 1500 kJ mol^{-1}: $CaCl_2$ is much more exothermic than CaCl. Removing a third electron would mean breaking into the full inner shell of calcium, which costs 4912 kJ mol^{-1}. The extra lattice energy cannot repay that, so $CaCl_3$ would be strongly endothermic. Even CaCl, though exothermic, would not survive: $2CaCl(s) \rightarrow Ca(s) + CaCl_2(s)$ releases about 440 kJ mol^{-1}.

What makes a lattice energy large?

Lattice energy comes from the electrostatic attraction between oppositely charged ions. That attraction is stronger when the charges are larger and when the ions can get closer together. So lattice energy becomes more exothermic when:

- **the ions are smaller:** their centres are closer, so they attract more strongly. Lattice energies become less negative down a group.
- **the ions have larger charges:** the attraction depends on the product of the charges, so going from 1+ and 1− ions to 2+ and 2− ions roughly quadruples it.

Compound	LE / kJ mol^{-1}	What it shows
NaF, NaCl, NaBr, NaI	−928, −787, −751, −703	bigger anion, less exothermic
NaCl, KCl	−787, −718	bigger cation, less exothermic
$MgCl_2$, $CaCl_2$	−2524, −2255	2+ ions give much larger values; Mg^{2+} is smaller than Ca^{2+}
NaF, MgO	−928, −3845	similar-sized ions, but charges 1+/1− against 2+/2−: about four times larger

Theoretical and experimental lattice energies

A lattice energy can also be calculated purely from theory, using the **perfect ionic model**: the ions are treated as perfect spheres with their charge spread evenly, held together only by electrostatic attraction. Comparing this theoretical value with the experimental value from a Born–Haber cycle tests whether the model is right.



purely ionic: both ions spherical

polarised: electron density pulled towards the cation (covalent character

A small, highly charged cation distorts the electron cloud of a large anion, so some electron density is shared between the ions.

- If the two values agree closely, the compound is almost **purely ionic**.

- If the Born–Haber value is **more exothermic** than the theoretical one, the bonding has some **covalent character**. The cation pulls the anion's electron cloud towards itself (**polarisation**), concentrating electron density between the ions; this extra attraction makes the real lattice stronger than the model predicts.
- **Polarising power** of a cation is greatest when it is small and highly charged (high charge density): Mg^{2+} polarises more than Na^{+}, and Al^{3+} more still.
- **Polarisability** of an anion is greatest when it is large (and highly charged): its outer electrons are far from the nucleus and easily pulled. I^{-} is far more polarisable than F^{-}.

Compound	Born–Haber LE / kJ mol^{-1}	Theoretical LE / kJ mol^{-1}	Difference
NaCl	about −787	about −770	about 2%: nearly purely ionic
AgI	about −890	about −780	about 12%: large, polarisable iodide
MgI_2	about −2330	about −1940	about 17%: small 2+ cation and large anion

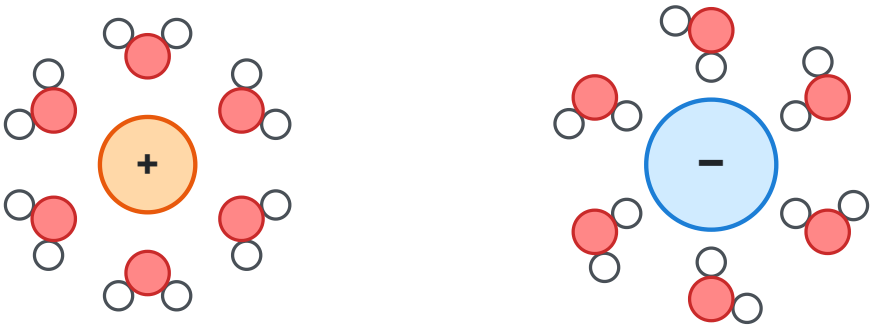
Enthalpies of solution and hydration

DEFINITION

The **enthalpy change of solution**, $\Delta_{sol}H$, is the enthalpy change when 1 mol of an ionic solid dissolves in enough water that the ions are far apart and do not interact: $NaCl(s) + aq \rightarrow Na^{+}(aq) + Cl^{-}(aq)$.

The **enthalpy change of hydration**, $\Delta_{hyd}H$, is the enthalpy change when 1 mol of gaseous ions is dissolved in enough water that further dilution causes no further enthalpy change: $Na^{+}(g) + aq \rightarrow Na^{+}(aq)$.

Water molecules are polar. The δ− oxygen atoms cluster round positive ions and the δ+ hydrogen atoms point at negative ions. These ion–dipole attractions release energy, so **hydration enthalpies are always negative**.



δ− oxygen atoms face the cation

δ+ hydrogen atoms face the anion

Hydrated ions. Oxygen atoms are red and hydrogen atoms white.

Hydration enthalpy is more exothermic for ions with a higher **charge density**: a smaller radius or a larger charge lets the water molecules get closer and attract more strongly.

Ion	Li^+	Na^+	K^+	Mg^{2+}	Ca^{2+}	F^-	Cl^-	Br^-	I^-
$\Delta_{hyd}H$ / kJ mol^{-1}	−519	−406	−322	−1920	−1650	−506	−364	−335	−293

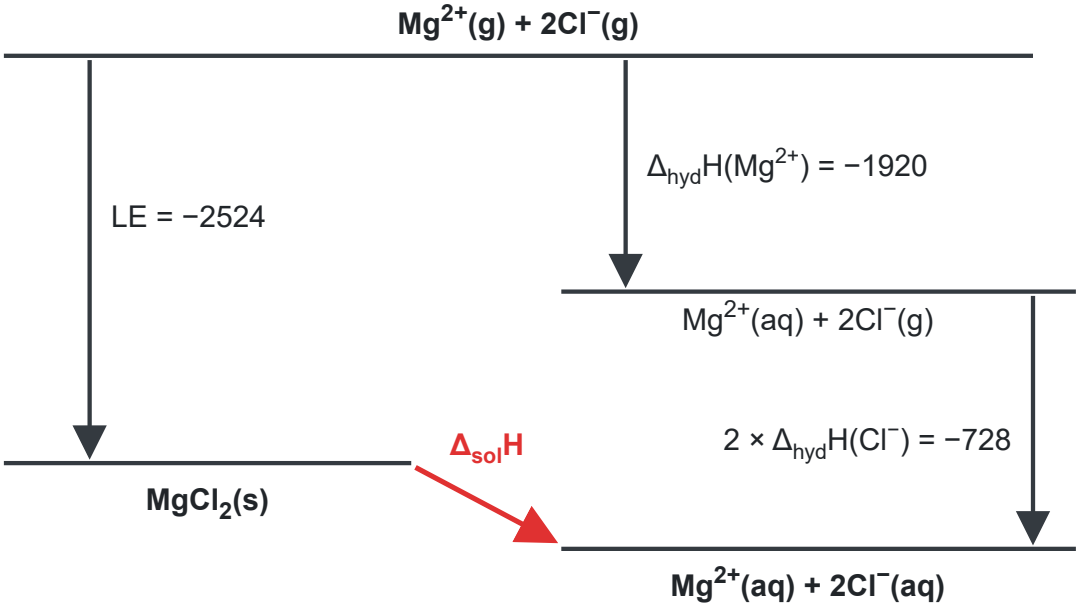
The solution cycle

Dissolving can be imagined in two stages: break the lattice into gaseous ions (the reverse of lattice formation, so +|LE|), then hydrate the ions. Hess's law gives:

Enthalpy change of solution

$$\Delta_{\text{sol}}H = -\text{LE} + \sum \Delta_{\text{hyd}}H$$

LE: lattice energy (negative, for formation) $\sum \Delta_{\text{hyd}}H$: hydration enthalpies of all the ions



Solution cycle for magnesium chloride, values in kJ mol^{-1} (not to scale). The direct route (red) equals the route up through the gaseous ions and down through hydration.

Enthalpy change of solution of magnesium chloride

1. Data / kJ mol^{-1}: LE($MgCl_2$) = −2524; $\Delta_{hyd}H(Mg^{2+})$ = −1920; $\Delta_{hyd}H(Cl^-)$ = −364.
2. Breaking up the lattice is the reverse of lattice formation: +2524 kJ mol^{-1}.
3. Hydrate one Mg^{2+} ion and **two** Cl^- ions:

$$\sum \Delta_{\text{hyd}}H = -1920 + 2(-364) = -2648\ \text{kJ mol}^{-1}$$

4. Add the two stages:

$$\Delta_{\text{sol}}H = +2524 + (-2648) = -124\ \text{kJ mol}^{-1}$$

Exothermic: the hydration of the ions releases more energy than breaking the lattice takes in, and the solution warms up as magnesium chloride dissolves.

Lattice energy and total hydration enthalpy are usually large and similar in size, so $\Delta_{sol}H$, the small difference between them, can come out either side of zero. A very endothermic $\Delta_{sol}H$, where the lattice is far stronger than the hydration can repay, usually means the compound is insoluble. But enthalpy is not the whole story: ammonium

nitrate dissolves readily even though $\Delta_{sol}H$ is +26 kJ mol^{-1} and the solution goes cold. To see why, we need entropy.

Entropy

A **spontaneous** (feasible) change is one that happens on its own once started, without anything continuing to drive it: a gas spreading through a room, ice melting on a warm day, an acid neutralising an alkali. Most feasible reactions are exothermic, but not all. Ice melting and ammonium nitrate dissolving both take in heat. The quantity that decides the direction of every change is **entropy**, S.

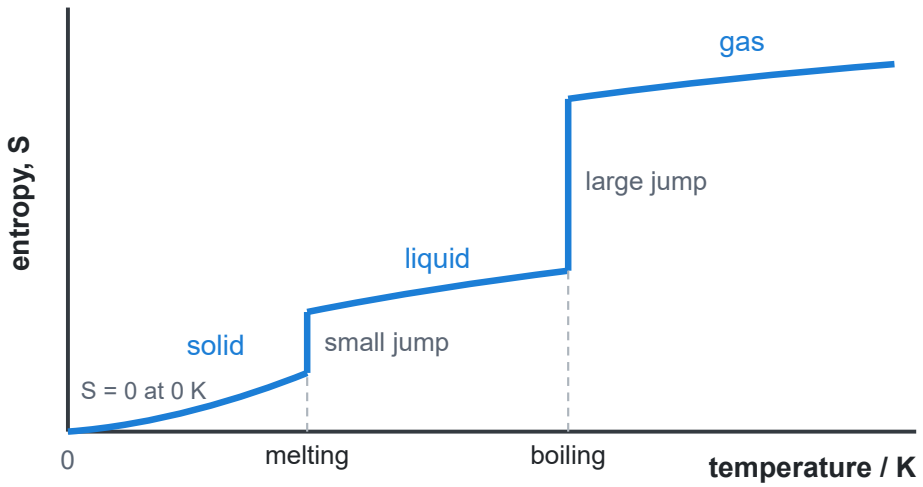
DEFINITION

Entropy is a measure of the number of ways that particles, and the quanta of energy they hold, can be arranged. The more ways there are, the greater the entropy, and the more disordered the system appears.

Energy comes in small packets (quanta) that particles can swap among themselves. With more quanta, or more particles to share them, the number of possible arrangements grows enormously. A system left to itself drifts into the most probable state: the one with the most arrangements, which is why energy spreads out and never concentrates by itself.

Entropy of substances

- **Gases** have far higher entropies than liquids, and liquids higher than solids, because their particles are free to move and can be arranged in vastly more ways.
- Entropy **increases with temperature**: more energy quanta means more ways to arrange them, and particles vibrate and move more.
- There is a sudden jump at each change of state. The jump at boiling is much bigger than at melting, because a gas is so much more disordered than a liquid.
- More complex molecules, with more atoms that can move in more ways, have higher entropies than simple ones; mixtures have higher entropies than pure substances.
- Only a **perfect crystal at 0 K** has zero entropy. Every substance at 298 K, elements included, has a positive standard entropy, $S^{\ominus}$, measured in **J K^{-1} mol^{-1}** (note: joules, not kilojoules).



Entropy rises with temperature, with a small jump on melting and a much larger one on boiling.

Predicting the sign of ΔS_{system}

The **system** is the reacting chemicals; everything else (the flask, the water, the air) is the **surroundings**. You can usually predict the sign of the system's entropy change from the equation. ΔS_{system} is positive when a solid or liquid turns into a gas, and when the number of moles of gas increases; it is negative when the reverse happens.

- $NH_4Cl(s) \rightarrow NH_3(g) + HCl(g)$: a solid becomes two gases, so ΔS_{system} is large and positive.
- $2Mg(s) + O_2(g) \rightarrow 2MgO(s)$: a gas is used up to make a solid, so ΔS_{system} is negative.
- $N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$: four moles of gas become two, so ΔS_{system} is negative.
- $NaCl(s) + aq \rightarrow Na^+(aq) + Cl^-(aq)$: an ordered lattice breaks into freely moving ions, so ΔS_{system} is usually positive.

Calculating ΔS_{system}

Entropy change of the system

$$\Delta S^\circ_{\text{system}} = \sum S^\circ_{\text{products}} - \sum S^\circ_{\text{reactants}}$$

Multiply each standard entropy by its number of moles in the equation Units: J K^{-1} mol^{-1}

ΔS for $N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$

1. Data / J K^{-1} mol^{-1}: $S^\ominus[N_2(g)] = 191.6$; $S^\ominus[H_2(g)] = 130.6$; $S^\ominus[NH_3(g)] = 192.3$.

2. Products: 2 × 192.3 = 384.6. Reactants: 191.6 + 3 × 130.6 = 583.4.

3. $$\Delta S_{\text{system}} = 384.6 - 583.4 = -198.8 \text{ J K}^{-1} \text{ mol}^{-1}$$

Negative, as predicted: four moles of gas become two.

Entropy of the surroundings and total entropy

An exothermic reaction releases energy into the surroundings. Those extra quanta can be shared among the surrounding particles in many more ways, so the entropy of the surroundings **increases**. An endothermic reaction takes energy from the surroundings and lowers their entropy. The effect is bigger when the surroundings are cold: adding a given amount of energy to something cold makes a larger relative difference to the number of ways it can be shared than adding it to something already hot.

Entropy change of the surroundings and total entropy change

$$\Delta S_{\text{surroundings}} = -\frac{\Delta H}{T} \qquad \Delta S_{\text{total}} = \Delta S_{\text{system}} + \Delta S_{\text{surroundings}}$$

ΔH in J mol⁻¹ (× 1000 from kJ) T in K Entropies in J K⁻¹ mol⁻¹

KEY IDEA

A change is **feasible** (thermodynamically possible) if **ΔS_{total} is positive**. This is the second law of thermodynamics: the total entropy of the system and its surroundings increases in every spontaneous change.

Is decomposing calcium carbonate feasible at 298 K?

1. $CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$. Data: $\Delta_f H$ / kJ mol^{-1}: $CaCO_3$ −1206.9, CaO −635.1, CO_2 −393.5. $S^{\ominus}$ / J K^{-1} mol^{-1}: $CaCO_3$ 92.9, CaO 39.7, CO_2 213.6.

2. Enthalpy change: $\Delta H = (-635.1 - 393.5) - (-1206.9) = +178.3$ kJ mol^{-1}.

3. Entropy change of the system: $\Delta S_{system} = (39.7 + 213.6) - 92.9 = +160.4$ J K^{-1} mol^{-1}.

Positive, because a gas is made from a solid.

4. Entropy change of the surroundings at 298 K, converting ΔH to J mol^{-1}:

$$\Delta S_{\text{surroundings}} = -\frac{178\,300}{298} = -598.3\ \text{J K}^{-1}\,\text{mol}^{-1}$$

5. Total:

$$\Delta S_{\text{total}} = +160.4 + (-598.3) = -437.9\ \text{J K}^{-1}\,\text{mol}^{-1}$$

Negative, so limestone does not decompose at room temperature.

6. At what temperature does it become feasible? At the boundary $\Delta S_{\text{total}} = 0$, so $\Delta S_{\text{system}} = \Delta H/T$:

$$T = \frac{\Delta H}{\Delta S_{\text{system}}} = \frac{178\,300}{160.4} = 1110\ \text{K}$$

Above about 1110 K (840 °C), $\Delta S_{\text{surroundings}}$ is small enough for the positive ΔS_{system} to win. Lime kilns run hotter than this.

Changes of state

At its melting or boiling temperature, a substance is in equilibrium between the two states, and $\Delta S_{\text{total}} = 0$. That makes $T = \Delta H/\Delta S_{\text{system}}$ a way to calculate the temperature of a change of state. For water boiling, ΔH = +40.7 kJ mol^{-1} and ΔS_{system} = +109 J K^{-1} mol^{-1}, so T = 40 700 / 109 = 373 K. Below 373 K, $\Delta S_{\text{surroundings}}$ is more negative than ΔS_{system} is positive, and liquid water is the stable state; above it, the reverse.

How temperature affects feasibility

Since $\Delta S_{\text{total}} = \Delta S_{\text{system}} - \Delta H/T$, raising the temperature shrinks the $-\Delta H/T$ term. Whether that helps depends on the signs:

ΔH	ΔS_{system}	Feasible?	Example
− (exothermic)	+	at every temperature	$2H_2O_2(l) \rightarrow 2H_2O(l) + O_2(g)$
− (exothermic)	−	only **below** T = ΔH/ΔS	$N_2 + 3H_2 \rightarrow 2NH_3$, below about 464 K
+ (endothermic)	+	only **above** T = ΔH/ΔS	$CaCO_3 \rightarrow CaO + CO_2$, above about 1110 K
+ (endothermic)	−	never	$N_2(g) + 2O_2(g) \rightarrow 2NO_2(g)$

If ΔH is close to zero, −ΔH/T is small at any temperature, and feasibility depends almost entirely on ΔS_{system}.

Gibbs energy

Chemists usually combine ΔH and ΔS_{system} into one quantity that refers only to the system: the **Gibbs energy change**, ΔG.

From ΔS_total to ΔG

1. Start from the total entropy change:

$$\Delta S_{\text{total}} = \Delta S_{\text{system}} - \frac{\Delta H}{T}$$

2. Multiply every term by −T:

$$-T\Delta S_{\text{total}} = \Delta H - T\Delta S_{\text{system}}$$

3. Define the right-hand side as the Gibbs energy change:

$$\Delta G = \Delta H - T\Delta S_{\text{system}} = -T\Delta S_{\text{total}}$$

Because T is always positive, ΔG is negative exactly when ΔS_{total} is positive.

Gibbs energy change

$$\Delta G = \Delta H - T\Delta S_{\text{system}}$$

ΔG, ΔH in kJ mol⁻¹ ΔS in kJ K⁻¹ mol⁻¹ (÷ 1000 from J) T in K

KEY IDEA

A reaction is **feasible when ΔG is negative**. When ΔG = 0 the system is at the boundary, in equilibrium. A graph of ΔG against T is a straight line with intercept ΔH and gradient $-\Delta S_{system}$, and it crosses zero at T = $\Delta H/\Delta S_{system}$.

ΔG for decomposing calcium carbonate

1. $\Delta H = +178.3$ kJ mol^{-1} and $\Delta S_{system} = +160.4$ J K^{-1} mol^{-1} = +0.1604 kJ K^{-1} mol^{-1}.

2. At 298 K:

$$\Delta G = 178.3 - 298 \times 0.1604 = +130.5 \text{ kJ mol}^{-1}$$

Positive: not feasible, as the entropy calculation showed.

3. At 1200 K:

$$\Delta G = 178.3 - 1200 \times 0.1604 = -14.2 \text{ kJ mol}^{-1}$$

Negative: feasible at kiln temperatures.

Gibbs energy and equilibrium constants

ΔG is linked to the equilibrium constant K (K_c or K_p) of the reaction by

Gibbs energy and the equilibrium constant

$$\Delta G = -RT \ln K$$

R = 8.31 J K⁻¹ mol⁻¹, so ΔG here is in J mol⁻¹ T in K

A negative ΔG means ln K is positive, so K > 1 and products predominate at equilibrium. The more negative ΔG, the larger K. A reaction with ΔG of about −60 kJ mol^{-1} at 298 K has K of about 10^{10}: it goes essentially to completion. A positive ΔG gives K < 1, but not zero, so a little product still forms.

K	below 10^{-10}	about 0.1	1	about 10	above 10^{10}
Position of equilibrium	reaction effectively doesn't go	reactants predominate	similar amounts	products predominate	reaction effectively complete

Feasible does not mean fast

ΔG and ΔS_{total} say whether a reaction *can* happen, not how quickly. A mixture of hydrogen and oxygen at room temperature has a hugely negative ΔG for forming water, yet nothing happens until a spark supplies the activation energy. Diamond should turn into graphite at room temperature (ΔG is negative), but the rate is immeasurably slow. Such systems are **thermodynamically unstable but kinetically stable** (kinetically inert): a high activation energy

stops the feasible reaction from happening at a noticeable rate. Remember too that the calculations use standard data and assume ΔH and ΔS stay the same at other temperatures, which is only approximately true.

Entropy and solubility

Now return to dissolving. When an ionic solid dissolves, the ions leave an ordered lattice and spread through the solution, so ΔS_{system} is usually positive.

- If $\Delta_{sol}H$ is **negative**, $\Delta S_{surroundings}$ is positive too, so ΔS_{total} is positive and the solid dissolves at any temperature.
- If $\Delta_{sol}H$ is **positive**, $\Delta S_{surroundings}$ is negative. The solid still dissolves if the positive ΔS_{system} is bigger. Ammonium nitrate: $\Delta_{sol}H$ = +25.7 kJ mol^{-1}, ΔS_{system} = +108.7 J K^{-1} mol^{-1}. At 298 K, $\Delta S_{surroundings}$ = −25 700/298 = −86.2, so ΔS_{total} = +22.5 J K^{-1} mol^{-1}: it dissolves, and cold packs use exactly this.
- Raising the temperature makes the negative $-\Delta_{sol}H/T$ term smaller in size, so salts with an endothermic $\Delta_{sol}H$ generally become more soluble when heated.
- If $\Delta_{sol}H$ is very endothermic, because the lattice energy is far larger than the hydration enthalpies, $\Delta S_{surroundings}$ is too negative to overcome and the solid is insoluble.

Quick checks

QUICK CHECK

Which equation represents the lattice energy of magnesium oxide?

- $Mg(s) + ½O_2(g) \rightarrow MgO(s)$
- $MgO(s) \rightarrow Mg^{2+}(g) + O^{2-}(g)$
- ✓ $Mg^{2+}(g) + O^{2-}(g) \rightarrow MgO(s)$
- $Mg^{2+}(aq) + O^{2-}(aq) \rightarrow MgO(s)$

Lattice energy is the formation of 1 mol of solid from its gaseous ions. The first option is the enthalpy of formation; the second is the lattice breaking up (same size, opposite sign); aqueous ions are wrong because hydration energy would be involved.

QUICK CHECK

Use the data to find the first electron affinity of bromine. Data / kJ mol^{-1}: $\Delta_f H$(KBr) = −394; $\Delta_{at} H$(K) = +89; first IE(K) = +419; $\Delta_{at} H$(Br) = +112; LE(KBr) = −689.

- +325 kJ mol^{-1}
- −1703 kJ mol^{-1}
- −915 kJ mol^{-1}
- −325 kJ mol^{-1}

$\Delta_f H = \Delta_{at} H(K) + IE + \Delta_{at} H(Br) + EA + LE$, so EA = −394 − (89 + 419 + 112 − 689) = −394 − (−69) = −325 kJ mol^{-1}. A positive answer comes from a sign error with the lattice energy.

QUICK CHECK

The Born–Haber lattice energy of a compound is much more exothermic than its theoretical lattice energy. What does this show?

- The ions are larger than expected.
- The bonding has significant covalent character.
- The compound is purely ionic.
- The compound will be very soluble in water.

The theoretical value assumes perfectly spherical ions. A polarising cation distorts the anion, putting electron density between the ions; this covalent character strengthens the lattice beyond what the ionic model predicts.

QUICK CHECK

Which change has the largest positive ΔS_{system}?

- $H_2O(g) \rightarrow H_2O(l)$
- $N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$
- $CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$ ✓
- $Na(s) + \frac{1}{2}Cl_2(g) \rightarrow NaCl(s)$

Only the decomposition of calcium carbonate creates a gas from a solid. The other three all reduce the amount or freedom of gas particles, so their ΔS_{system} is negative.

QUICK CHECK

A reaction has $\Delta H = -120$ kJ mol^{-1} and $\Delta S_{system} = -150$ J K^{-1} mol^{-1}. Above what temperature is it no longer feasible?

- 0.8 K
- 800 K ✓
- 1250 K
- It is feasible at all temperatures.

Feasible while $\Delta G = \Delta H - T\Delta S$ is negative. $\Delta G = 0$ when $T = \Delta H/\Delta S = -120\,000 / -150 = 800$ K. Above that, $-T\Delta S$ (positive) outweighs the negative ΔH. Forgetting to convert kJ to J gives 0.8 K.

QUICK CHECK

Why is the hydration enthalpy of Mg^{2+} much more exothermic than that of Na^{+}?

- Mg^{2+} has more electrons than Na^{+}.
- Mg^{2+} forms covalent bonds with water.
- Mg^{2+} is smaller and has a larger charge, so it attracts the δ− oxygen of water molecules more strongly.
- Mg^{2+} is surrounded by hydrogen atoms of water molecules.

Hydration energy comes from ion–dipole attraction. Mg^{2+} has a much higher charge density than Na^{+} (twice the charge, smaller radius), so water's δ− oxygen atoms are pulled in closer and more strongly. Cations are surrounded by oxygen, not hydrogen, atoms.

QUICK CHECK

The conversion of diamond into graphite has a negative ΔG at room temperature, yet diamonds do not change. Why?

- ΔS_{total} for the change is negative.
- The activation energy is so high that the rate is negligible.
- The equilibrium constant is less than 1.
- Graphite has a lower entropy than diamond.

A negative ΔG means ΔS_{total} is positive and K > 1, so the change is feasible. Feasibility says nothing about rate: diamond is kinetically stable because the activation energy for rearranging its strong covalent network is enormous.